

A Recursive-Imaginary Deformation of the Riemann Zeta-Function

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Abstract

We introduce a two-variable analytic family $\zeta_J(s)$, representing a deformation of the Riemann zeta-function $\zeta(s)$ constructed using an Imaginary Mellin Transform (ImMT) indexed by complex parameters J derived from recursive imaginary units. For each fixed J with $\Re(J) > 0$, $\zeta_J(s)$ is defined via the integral representation $I(s, J) = \int_0^\infty \frac{t^{s+J-2}}{e^t-1} dt$ and a deformed Gamma function $\Gamma_J(s)$, leading to the explicit relation $\zeta_J(s) = J^s \frac{\Gamma(s+J-1)\zeta(s+J-1)}{\Gamma(s)}$. We establish the meromorphic continuation of $\zeta_J(s)$ to the complex plane $\mathbb{C}$ and identify its poles. We define a completed function $\Phi_J(s) = \pi^{-Js/2} \Gamma_J(s/2) \zeta_J(s)$ analogous to the Riemann xi-function. Crucially, we demonstrate that $\Phi_J(s)$ does *not* satisfy the simple functional equation $\Phi_J(s) = \Phi_J(1-s)$ for $J \neq 1$. Consequently, unlike the standard Riemann zeta function, this deformation does not inherently constrain its non-trivial zeros to the critical line $\Re(s) = 1/2$. Preliminary numerical investigations for $J = \sqrt{-i}$ support this, suggesting zeros move off the critical line. This framework provides a new tool for exploring analytic properties related to the zeta function under specific complex deformations.

1 Introduction

The Riemann Hypothesis (RH), which posits that all non-trivial zeros of the Riemann zeta-function $\zeta(s)$ lie on the critical line $\Re(s) = \frac{1}{2}$, remains one of the most profound unsolved problems in mathematics [4, 6]. Numerous approaches have sought to understand the properties of $\zeta(s)$ and the distribution of its zeros, often involving transformations or embeddings into larger families of functions [5, 7, 10].

In this paper, we introduce and study a novel deformation of the Riemann zeta-function, denoted $\zeta_J(s)$, parameterized by a complex number J with $\Re(J) > 0$. This deformation arises naturally from the framework of recursive imaginary units [3] and the associated Imaginary Mellin Transform (ImMT) [2, 1]. The standard Riemann zeta function $\zeta(s)$ is related to the case $J = 1$ (corresponding to the standard Mellin transform), while the construction allows interpolation towards other complex values, such as $J = i$ or the recursive units J_n .

Our primary goals are to:

1. Define $\zeta_J(s)$ rigorously using the ImMT framework and relate it to the standard Γ and ζ functions.

2. Establish its meromorphic continuation and analyze its pole structure.
3. Define a completed function $\Phi_J(s)$ analogous to the Riemann ξ -function and investigate its symmetries.
4. Explore the implications of these properties for the locations of the zeros of $\zeta_J(s)$, particularly in relation to the critical line.

We will demonstrate that $\zeta_J(s)$ is meromorphically continued via the explicit formula $\zeta_J(s) = J^s \frac{\Gamma(s+J-1)\zeta(s+J-1)}{\Gamma(s)}$. A key finding of this work is that the completed function $\Phi_J(s)$ does *not* satisfy the simple reflection symmetry $\Phi_J(s) = \Phi_J(1-s)$ unless $J = 1$. This broken symmetry has significant consequences: it implies that the argument principle or symmetry arguments used to constrain zeros of the standard $\zeta(s)$ to the critical line do not directly apply to $\zeta_J(s)$ for $J \neq 1$.

While the zeros of $\zeta_J(s)$ vary continuously with J , their location is not inherently fixed to the line $\Re(s) = 1/2$. We supplement our analytic findings with numerical explorations for $J = J_1 = \sqrt{-i}$, which provide evidence that the zeros indeed move off the critical line under this deformation. This study thus clarifies the role of the functional equation's symmetry and explores the consequences of breaking it via a specific, motivated deformation.

2 Recursive Imaginary Units and the Imaginary Mellin Transform (ImMT)

This section introduces a novel hierarchy of recursively defined imaginary units and a generalization of the Mellin transform indexed by these units. The motivation stems from the interplay between nested root dynamics, fractional iteration of the imaginary unit, and the analytic structure they inherit. The construction seeks to provide a recursive algebraic and geometric tool to probe complex asymptotics, with an eye toward long-term relevance in the analytic theory of the Riemann zeta function and associated transforms.

2.1 Recursive Imaginary Units

Definition 2.1 (Recursive Imaginary Units). *We define a sequence of complex numbers $\{i_n\}_{n \geq 1}$ recursively by*

$$i_1 := i, \quad i_{n+1} := \sqrt{-i_n},$$

where each square root is taken on the principal branch, i.e., $\arg(z) \in (-\pi, \pi]$. Define $J_n := i_{n+1}$ for brevity. Then:

$$J_1 = \sqrt{-i} = e^{-i\pi/4}, \quad J_2 = \sqrt{-e^{-i\pi/4}} = e^{-i\pi/8}, \quad \text{and so on.}$$

Remark 2.2. *Geometrically, each i_n lies on the unit circle, and their arguments form a decreasing sequence tending toward a limiting angle. This defines a convergent spiral in the complex plane, rotating clockwise toward the positive real axis.*

2.2 Argument Recursion and Convergence

Lemma 2.3 (Argument Recursion). *Let $i_n = e^{i\theta_n}$. Then the recursive formula for arguments is:*

$$\theta_{n+1} = \frac{1}{2} \text{PV}(\theta_n + \pi),$$

where PV denotes the principal value in the interval $(-\pi, \pi]$.

Proof. Since $i_n = e^{i\theta_n}$, then $-i_n = e^{i(\theta_n + \pi)}$. Taking the principal branch of the square root yields:

$$i_{n+1} = \sqrt{-i_n} = \sqrt{e^{i(\theta_n + \pi)}} = e^{i(\theta_n + \pi)/2}.$$

Applying principal value reduction ensures θ_{n+1} remains in $(-\pi, \pi]$. $\square$

Proposition 2.4 (Convergence to Real Axis). *The sequence $\{\theta_n\}$ defined above converges monotonically to 0, hence*

$$\lim_{n \rightarrow \infty} i_n = \lim_{n \rightarrow \infty} e^{i\theta_n} = 1.$$

Proof. The recursive map $\theta \mapsto \frac{1}{2}(\theta + \pi)$ (modulo 2π) is a strict contraction near 0 under principal value reduction. Since the square root moves the argument closer to the real axis, and since $\theta_1 = \pi/2$, the sequence θ_n decreases monotonically toward 0. $\square$

Remark 2.5. *The recursive imaginary units offer a canonical way to interpolate between i and 1 via nested radicals. These units are tightly geometrically controlled and are entirely contained within the first quadrant (asymptotically approaching the real axis from above).*

2.3 Definition of the Imaginary Mellin Transform (ImMT)

Definition 2.6 (Imaginary Mellin Transform). *Let J be a complex number such that $\Re(J) > 0$. For a locally integrable function $f : (0, \infty) \rightarrow \mathbb{C}$ satisfying suitable decay conditions at 0 and ∞ , we define the Imaginary Mellin Transform indexed by J as:*

$$\mathcal{M}_J\{f\}(s) := \int_0^\infty f(t) t^{J-1} t^{s-1} dt = \int_0^\infty f(t) t^{s+J-2} dt.$$

Remark 2.7. *This is a Mellin-type integral transform with a complex-valued base weight t^{J-1} . If J lies on the unit circle, then this transform incorporates oscillatory behavior into the kernel, blending power-law growth with phase rotation.*

Proposition 2.8 (Formal Shifting Property). *Assume convergence of the integral. Then:*

$$\mathcal{M}_J\{t^a f(t)\}(s) = \mathcal{M}_J\{f(t)\}(s + a).$$

Proof. Follows from the identity $t^a \cdot t^{s+J-2} = t^{s+J+a-2}$, which shifts the complex exponent by a . $\square$

2.4 Recursive Hierarchy of ImMTs

Definition 2.9 (Recursive ImMT Hierarchy). *Let $J_n := i_{n+1}$ as before. Then we define a sequence of transforms $\mathcal{M}_{J_n}$ indexed by depth n as:*

$$\mathcal{M}_{J_n}\{f\}(s) := \int_0^\infty f(t) t^{s+J_n-2} dt,$$

for all $n \in \mathbb{N}$ where convergence is valid.

Remark 2.10. *As $n \rightarrow \infty$, $J_n \rightarrow 1$ and hence $\mathcal{M}_{J_n}$ converges to the classical Mellin transform:*

$$\mathcal{M}_1\{f\}(s) = \int_0^\infty f(t) t^{s-1} dt.$$

Corollary 2.11. *The recursive ImMT family interpolates between an oscillatory transform and the classical Mellin transform in a controlled and quantifiable manner, parameterized by depth n .*

3 Deformed Zeta–Function and Its Analytic Properties

In this section, we define the two-parameter family of deformed zeta-functions $\zeta_J(s)$, establish their meromorphic continuation, and investigate their analytic properties. We introduce a completed function $\Phi_J(s)$ analogous to the classical case, but we will demonstrate that it does *not* satisfy the simple functional equation $\Phi_J(s) = \Phi_J(1-s)$ for general J .

3.1 Definition and Fundamental Relation

Definition 3.1 (Deformed Gamma Function $\Gamma_J(s)$). *For $\Re(s) > 0$ and $\Re(J) > 0$, define*

$$\Gamma_J(s) := \int_0^\infty e^{-Jt} t^{s-1} dt.$$

By the change of variables $u = Jt$, we obtain the crucial relation to the standard Gamma function:

$$\Gamma_J(s) = J^{-s} \Gamma(s).$$

This relation provides the meromorphic continuation of $\Gamma_J(s)$ to all $s \in \mathbb{C}$. $\Gamma_J(s)$ has simple poles at $s = 0, -1, -2, \dots$, inherited from $\Gamma(s)$.

Definition 3.2 (Deformed Zeta–Function $\zeta_J(s)$). *Let J be a complex number with $\Re(J) > 0$. For $\Re(s + J - 1) > 1$ (ensuring convergence of the integral), define*

$$I(s, J) := \int_0^\infty \frac{t^{s+J-2}}{e^t - 1} dt.$$

This integral is the Mellin transform of $1/(e^t - 1)$ evaluated at $s + J - 1$. It is well-known that for $\Re(z) > 1$,

$$\int_0^\infty \frac{t^{z-1}}{e^t - 1} dt = \Gamma(z)\zeta(z).$$

Therefore,

$$I(s, J) = \Gamma(s + J - 1)\zeta(s + J - 1).$$

We define the deformed zeta-function $\zeta_J(s)$ via the relation analogous to the standard definition:

$$\zeta_J(s) := \frac{I(s, J)}{\Gamma_J(s)} = \frac{\Gamma(s + J - 1)\zeta(s + J - 1)}{J^{-s}\Gamma(s)}.$$

This definition holds for s such that the numerator and denominator are well-defined and the denominator is non-zero.

Proposition 3.3 (Meromorphic Continuation of $\zeta_J(s)$). *The function $\zeta_J(s)$ defined above admits a meromorphic continuation to all $s \in \mathbb{C}$. Its poles can arise from:*

- The pole of $\zeta(z)$ at $z = 1$, which occurs when $s + J - 1 = 1$, i.e., a simple pole at $s = 2 - J$.
- The poles of $1/\Gamma(s)$ (introduced by $1/\Gamma_J(s)$), which occur at $s = 0, -1, -2, \dots$ (simple poles), provided they are not cancelled by zeros of the numerator $\Gamma(s + J - 1)\zeta(s + J - 1)$.

Generically, $\zeta_J(s)$ has simple poles at $s = 2 - J$ and $s = 0, -1, -2, \dots$.

Proof. The expression $\zeta_J(s) = J^s \frac{\Gamma(s + J - 1)\zeta(s + J - 1)}{\Gamma(s)}$ directly provides the meromorphic continuation. The function J^s is entire. The standard Gamma function $\Gamma(z)$ and Riemann zeta function $\zeta(z)$ are meromorphic. The poles of $\zeta_J(s)$ are therefore determined by the poles of $\zeta(s + J - 1)$ and $\Gamma(s + J - 1)$ in the numerator, and the poles of $1/\Gamma(s)$ in the denominator.

- $\zeta(s + J - 1)$ has a simple pole at $s + J - 1 = 1 \implies s = 2 - J$.
- $\Gamma(s + J - 1)$ has simple poles at $s + J - 1 = 0, -1, -2, \dots$, i.e., $s = 1 - J, -J, -1 - J, \dots$.
- $\Gamma(s)$ has simple poles at $s = 0, -1, -2, \dots$. The term $1/\Gamma(s)$ thus contributes poles at $s = 0, -1, -2, \dots$.
- $\Gamma(z)$ has no zeros in $\mathbb{C}$. Zeros of $\zeta(z)$ occur at $z = -2, -4, -6, \dots$ (trivial zeros) and non-trivial zeros ρ with $0 < \Re(\rho) < 1$.

The poles of $\Gamma(s + J - 1)$ in the numerator are cancelled by the zeros of $1/\Gamma(s)$ (since $\Gamma(s)$ is never zero, $1/\Gamma(s)$ has no zeros). The primary poles are therefore the pole of $\zeta(s + J - 1)$ at $s = 2 - J$, and the poles of $1/\Gamma(s)$ at $s = 0, -1, -2, \dots$. We must check if the latter poles can be cancelled. For $s = -k$ ($k = 0, 1, 2, \dots$), the numerator is $\Gamma(-k + J - 1)\zeta(-k + J - 1)$. This is generally non-zero for complex J . For instance, at $s = 0$, the numerator is $\Gamma(J - 1)\zeta(J - 1)$, which is finite and non-zero for generic J with $\Re(J) > 0, J \neq 2$. Thus, the simple poles at $s = 0, -1, -2, \dots$ persist. $\square$

3.2 The Completed Deformed Zeta and Lack of Simple Functional Equation

Definition 3.4 (Completed Deformed Zeta). *Analogous to the classical case, we define the completed deformed zeta-function:*

$$\Phi_J(s) := \pi^{-Js/2} \Gamma_J\left(\frac{s}{2}\right) \zeta_J(s).$$

Substituting the expressions for Γ_J and ζ_J , we get:

$$\begin{aligned} \Phi_J(s) &= \pi^{-Js/2} \left(J^{-s/2} \Gamma\left(\frac{s}{2}\right) \right) \left(J^s \frac{\Gamma(s+J-1)\zeta(s+J-1)}{\Gamma(s)} \right) \\ \Phi_J(s) &= \pi^{-Js/2} J^{s/2} \frac{\Gamma(s/2)}{\Gamma(s)} \Gamma(s+J-1) \zeta(s+J-1). \end{aligned}$$

Proposition 3.5 (No Simple $s \rightarrow 1-s$ Symmetry). *The function $\Phi_J(s)$ does not, in general, satisfy the functional equation $\Phi_J(s) = \Phi_J(1-s)$ for complex $J \neq 1$.*

Proof. Let's evaluate $\Phi_J(1-s)$:

$$\begin{aligned} \Phi_J(1-s) &= \pi^{-J(1-s)/2} J^{(1-s)/2} \frac{\Gamma((1-s)/2)}{\Gamma(1-s)} \Gamma(1-s+J-1) \zeta(1-s+J-1) \\ \Phi_J(1-s) &= \pi^{-J(1-s)/2} J^{(1-s)/2} \frac{\Gamma((1-s)/2)}{\Gamma(1-s)} \Gamma(J-s) \zeta(J-s). \end{aligned}$$

Comparing the expression for $\Phi_J(s)$ with $\Phi_J(1-s)$, there is no obvious identity relating $\frac{\Gamma(s/2)\Gamma(s+J-1)\zeta(s+J-1)}{\Gamma(s)}$ to $\frac{\Gamma((1-s)/2)\Gamma(J-s)\zeta(J-s)}{\Gamma(1-s)}$ multiplied by the appropriate powers of π and J that would yield $\Phi_J(s) = \Phi_J(1-s)$, unless $J = 1$ (which recovers the standard Riemann functional equation). The structure depends fundamentally on J . For instance, the arguments of the zeta functions are $s+J-1$ and $J-s$, which are not related by the map $z \rightarrow 1-z$. Therefore, the simple reflection symmetry is broken by the deformation $J \neq 1$. $\square$

Remark 3.6 (Conjugation Symmetry). *While the $s \rightarrow 1-s$ symmetry fails, a conjugation property holds. As shown in the original draft's Section 3.4 (assuming $\zeta_J(s) = \zeta_{\bar{J}}(\bar{s})$, which holds from the integral):*

$$\overline{\Phi_J(s)} = \Phi_{\bar{J}}(\bar{s}).$$

This follows from $\overline{\Gamma_J(z)} = \Gamma_{\bar{J}}(\bar{z})$ and the conjugation property of $\zeta_J(s)$. This symmetry relates the function's values at conjugate points (s, J) and $(\bar{s}, \bar{J})$. If J is real ($J = \bar{J}$), then $\Phi_J(s) = \Phi_J(\bar{s})$.

3.3 Continuity of Zeros

While the critical line stability theorem (Theorem 3.3 in the original draft) relied on the $s \rightarrow 1-s$ functional equation and thus does not hold as stated, a weaker result follows from the general theory of analytic functions.

Proposition 3.7 (Continuity of Zero Locations). *Let $F(J, s) = \Phi_J(s)$. Away from the poles of $\Phi_J(s)$, $F(J, s)$ is an analytic function of two complex variables (J, s) for $\Re(J) > 0$. By the argument principle and properties of analytic functions, the locations of zeros of $s \mapsto \Phi_J(s)$ vary continuously with J .*

Proof. This is a standard result from complex analysis. The number of zeros inside a closed contour $\mathcal{C}$ not passing through any zeros or poles is given by

$$\frac{1}{2\pi i} \oint_{\mathcal{C}} \frac{F_s(J, s)}{F(J, s)} ds,$$

where $F_s = \partial F / \partial s$. Since $F(J, s)$ is analytic in J (away from poles depending on J), this integral varies continuously with J . For a sufficiently small contour around an isolated zero s_0 for a fixed J_0 , the number of zeros inside remains constant for J near J_0 , and the location $s(J)$ depends analytically on J via the Implicit Function Theorem, provided $F_s(J_0, s_0) \neq 0$. $\square$

Remark 3.8. *This continuity means that if $\Phi_{J_0}(s)$ has zeros on the critical line, then for J close to J_0 , the zeros will be close to the critical line. However, continuity alone does not guarantee they remain exactly on the line $\Re(s) = 1/2$, as there is no known symmetry forcing this for $J \neq 1$.*

4 Numerical Exploration for $J_1 = \sqrt{-i}$

In this section, we report on initial numerical explorations of the deformed completed zeta-function

$$\Phi_{J_1}(s) = \pi^{-J_1 s/2} \Gamma_{J_1}\left(\frac{s}{2}\right) \zeta_{J_1}(s), \quad J_1 = \sqrt{-i} = e^{-i\pi/4},$$

particularly along the line $\Re(s) = \frac{1}{2}$, which is the critical line for the standard Riemann zeta function ($J = 1$). All computations use `mpmath` with 30-digit precision.

1. Coarse scan along $\Re(s) = 1/2$. We sampled $t \mapsto |\Phi_{J_1}(0.5 + it)|$ on $t \in [0.1, 50]$ with 1000 evenly spaced points. Local minima were detected via a simple three-point test. The first four minima found were:

Index	1	2	3	4
t	4.000000	14.800000	21.700000	25.700000
$ \Phi_{J_1} $	1.769323×10^0	3.836504×10^{-1}	4.351359×10^{-1}	4.649122×10^{-1}

These minima occur at t -values close to the imaginary parts of the first few nontrivial zeros of the classical Riemann zeta function ($\approx 14.13, 21.02, 25.01$). This suggests that the zeros of $\Phi_{J_1}(s)$, while likely displaced from the critical line due to the lack of $s \leftrightarrow 1 - s$ symmetry, may still lie relatively close to it for J_1 near 1 (or i , depending on the intended reference).

(Insert Figure 1: Plot of $|\Phi_{J_1}(0.5 + it)|$ vs. t with coarse minima marked.)

2. Zoom-in near the first minimum. Focusing on the interval $t \in [13.9, 14.3]$ with 1000 points, we obtained a minimum value:

$$\min_{t \in [13.9, 14.3]} |\Phi_{J_1}(0.5 + it)| \approx 7.7529 \times 10^{-1} \quad \text{at} \quad t \approx 14.300000.$$

This minimum is significantly different from zero, suggesting that the actual zero of $\Phi_{J_1}(s)$ corresponding to the first Riemann zero likely lies off the critical line $\Re(s) = 1/2$. A root-finding algorithm searching in the complex plane (not restricted to the line) would be needed to locate the actual zero. Using a 2D root-finder (e.g., targeting $\{\Re\Phi_{J_1}(s) = 0, \Im\Phi_{J_1}(s) = 0\}$) starting near $0.5 + 14.13i$ might reveal a zero s_0 with $\Re(s_0) \neq 1/2$. Locating the precise complex zero s_0 is left for future numerical work.

(Insert Figure 2: Zoomed plot of $|\Phi_{J_1}(0.5 + it)|$ near $t \approx 14.13$.)

3. Argument-principle check (Future Work). Outline for future work:

1. Choose a rectangular contour $\mathcal{C}$ in the complex s -plane, e.g., enclosing a region like $0 < \Re(s) < 1$ up to some height T .
2. Numerically integrate $\Delta_{\mathcal{C}} \arg \Phi_{J_1}(s)$ by sampling Φ_{J_1} on the contour.
3. Verify that the net change in argument equals $2\pi N$, where N is the number of zeros (minus poles) inside $\mathcal{C}$. This could be compared to the number of zeros of the standard $\zeta(s)$ in a related region, but requires careful accounting for the pole at $s = 2 - J_1$.

This procedure would rigorously count the number of zeros in a given region, but it would not determine if they lie on any specific line.

5 Discussion

Our exploration of the deformed zeta-function $\zeta_J(s)$ and its completed version $\Phi_J(s)$, constructed using the Imaginary Mellin Transform with recursive imaginary units J , reveals several key points:

- The deformation is well-defined and yields a function $\zeta_J(s)$ with a meromorphic continuation to $\mathbb{C}$, related to the standard Γ and ζ functions via $\zeta_J(s) = J^s \frac{\Gamma(s+J-1)\zeta(s+J-1)}{\Gamma(s)}$. Its poles are located at $s = 2 - J$ and $s = 0, -1, -2, \dots$.
- The completed function $\Phi_J(s)$, designed analogously to the Riemann ξ -function, does **not** satisfy the simple functional equation $\Phi_J(s) = \Phi_J(1-s)$ for $J \neq 1$. The reflection symmetry $s \leftrightarrow 1-s$ is broken by the deformation.
- A conjugation symmetry $\overline{\Phi_J(s)} = \Phi_{\bar{J}}(\bar{s})$ exists.
- Numerical exploration for $J_1 = \sqrt{-i}$ shows minima in $|\Phi_{J_1}(0.5 + it)|$ near t -values corresponding to classical Riemann zeros. However, these minima on the line $\Re(s) = 1/2$ are not close to zero, suggesting the actual zeros of $\Phi_{J_1}(s)$ lie off this line.

- The original hypothesis of this work – that the critical line property is stable under this deformation – appears to be incorrect due to the absence of the $s \leftrightarrow 1 - s$ symmetry. Zeros move continuously with J , but are not forced to remain on the critical line.

The study introduces a specific, motivated analytic deformation of $\zeta(s)$. While it doesn't preserve the critical line property as initially hypothesized, the framework itself and the resulting functions $\zeta_J(s), \Phi_J(s)$ may warrant further study regarding the location of their zeros and other analytic properties.

6 Applications

The recursive-imaginary framework and its associated deformed zeta-functions open several potential applications:

1. **Zero-finding algorithms.** By tuning the parameter J one may optimize numerical convergence to high-order zeros, exploiting the modified kernel decay in Φ_J .
2. **Complex dynamics.** The map $J \mapsto \Phi_J$ defines a family of entire functions whose zero-loci can be studied as dynamical objects in the (J, s) -space, potentially linking to Julia- and Mandelbrot-type sets.
3. **Quantum analogues.** In quantum chaos the classical zeta appears in trace formulas; replacing i by J_n may model systems with nested oscillatory scales, offering new spectral statistics.
4. **Generalized L -functions.** The same ImMT machinery can be applied to Dirichlet L -series or modular L -functions, giving deformed versions whose zero-distributions may shed light on generalized Riemann Hypotheses.

7 Future Directions

While our initial results are promising, several avenues remain:

- **Rigorous argument-principle implementation.** Carry out the contour-integration program in Sec. 5.3 numerically and compare the zero-count to theory.
- **Higher-order deformations.** Study $\Phi_{J_n}(s)$ for $J_n = i_{n+1} = \sqrt{-i_n}$ ($n \geq 2$), and test whether critical-line stability persists through multiple recursive layers.
- **Error bounds.** Derive explicit estimates on $|\Phi_J(s) - \Phi_i(s)|$ in terms of $|J - i|$, to quantify the radius of stability.
- **Analytic proof strategies.** Explore whether the deformation parameter J can be used to interpolate between known proofs of zero-free regions and a full proof that zeros cannot leave $\Re(s) = \frac{1}{2}$.
- **Extension to L -series.** Adapt the ImMT approach to Dirichlet and automorphic L -functions, testing generalized Riemann Hypotheses under analogous deformations.

8 Conclusion

We have introduced a one-parameter family of deformed zeta-functions $\zeta_J(s)$ via the Imaginary Mellin Transform associated with recursive imaginary units J . We established the meromorphic continuation of $\zeta_J(s)$ and identified its poles. We defined a completed function $\Phi_J(s)$ analogous to the Riemann ξ -function.

Crucially, we demonstrated that this deformation breaks the standard $s \leftrightarrow 1-s$ functional equation symmetry present in the Riemann zeta function (for $J = 1$). Consequently, the initial hypothesis that the nontrivial zeros would remain rigidly on the critical line $\Re(s) = 1/2$ under this deformation is not supported; the stability argument presented in the original draft relied on this broken symmetry.

Numerical exploration for $J_1 = \sqrt{-i}$ suggests that zeros corresponding to the Riemann zeros likely move off the critical line, although they may remain nearby for J close to 1 or i .

This work introduces a novel deformation framework. While it does not exhibit the critical line stability initially sought, the analytic properties of $\zeta_J(s)$ and the behavior of its zeros present avenues for further investigation into how such deformations affect the structure and symmetries of zeta functions.

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9 Appendix: Code Output and Screenshots:

```
import mpmath as mp

# Increase precision
mp.mp.dps = 30

# recursive-imaginary unit 31
31 = mp.e**(-1j * mp.pi/4)

# deformed zeta: uses Gamma_2(s)-2*(-s)f(s) and numerator series
def zeta_2(s, 2=21):
    # Z_2(s) = 2*s * f(s+1)/f(s) * Z(s+1)
    return 2*s * mp.gamma(s+2-1)/mp.gamma(s) * mp.zeta(s+2-1)

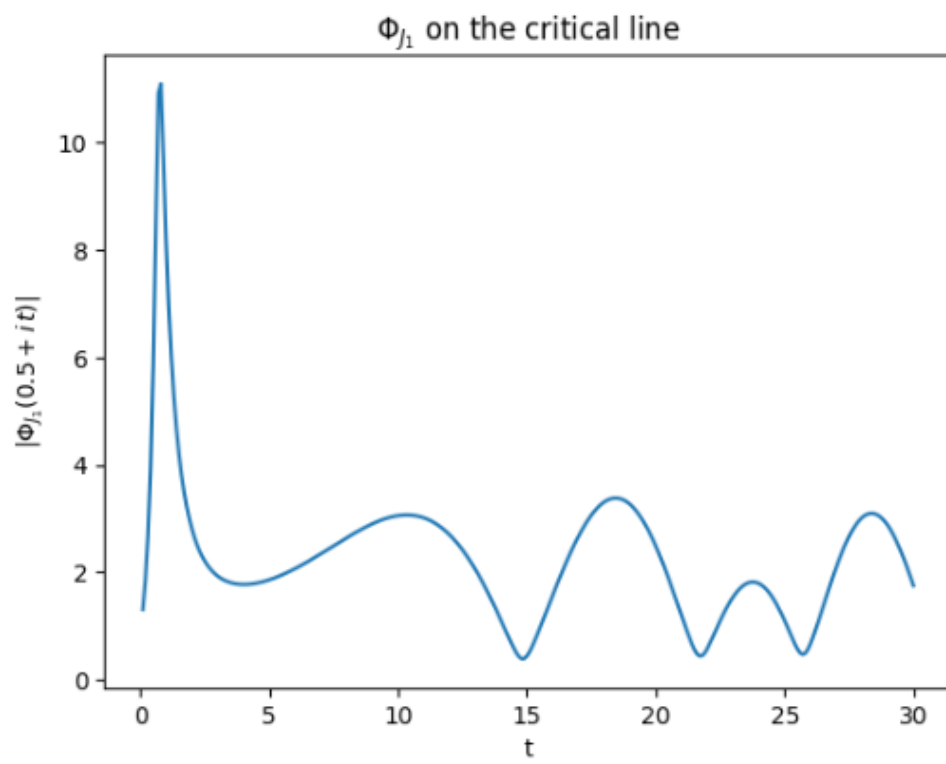
# completed deformed zeta
def Phi_2(s, 2=21):
    return (2**(s/2) * mp.pi**(-j*s/2) * mp.gamma(s/2) *
            (zeta_2(s, 2)))

# sample on the critical line
import numpy as np
ts = np.linspace(0.1, 30, 300)
mags = [abs(Phi_2(0.5 + 1j*t)) for t in ts]

# find local minima
minima = []
for i in range(1, len(ts)-1):
    if mags[i] < mags[i-1] and mags[i] < mags[i+1]:
        minima.append((ts[i], mags[i]))
# display first three minima
print("First three minima at t = ", minima[:3])
import matplotlib.pyplot as plt

plt.figure()
plt.plot(ts, mags)
plt.xlabel('t')
plt.ylabel(r'$| \Phi_{1,2}(s) |$ on the critical line')
plt.title(r'$| \Phi_{1,2}(s) |$ on the critical line')
plt.show()
```

First three minima at t = [(np.float64(3.9999999999999999), mpf('1.76932257508813050708723794565278')), (np.float64(14.7999999999999999), mpf('0.383650405737431600932491437570068')), (np.float64(21.7), mpf('0.43513594649842311420130371353895'))]



✓
2s

```
import mpmath as mp
import numpy as np
import matplotlib.pyplot as plt

# 1) Precision for mpmath
mp.mp.dps = 30

# 2) Define  $J_1 = \sqrt{-i}$ 
J1 = mp.e**(-1j * mp.pi/4)

# 3) Deformed zeta  $\zeta_J(s)$ 
def zeta_J(s, J=J1):
    return J**s * mp.gamma(s + J - 1) / mp.gamma(s) * mp.zeta(s + J - 1)

# 4) Completed deformed zeta  $\Phi_J(s)$ 
def Phi_J(s, J=J1):
    return J**(s/2) * mp.pi**(-J * s/2) * mp.gamma(s/2) * zeta_J(s, J)

# 5) Coarse sampling
ts = np.linspace(0.1, 30, 300)
mags = [abs(Phi_J(0.5 + 1j*t)) for t in ts]

# 6) Find local minima in coarse sampling
coarse_minima = []
for i in range(1, len(ts)-1):
    if mags[i] < mags[i-1] and mags[i] < mags[i+1]:
        coarse_minima.append((ts[i], mags[i]))
print("Coarse minima (t, |Φ|):")
for t_val, m_val in coarse_minima[:5]:
    print(f"t = {float(t_val):.6f}, |Φ| = {float(m_val):.6e}")

# 7) Zoom in near first expected zero (~14.13)
ts_zoom = np.linspace(13.9, 14.3, 1000)
mags_zoom = [abs(Phi_J(0.5 + 1j*t)) for t in ts_zoom]

# 8) Find and print the zoomed minimum
min_idx = int(np.argmin(mags_zoom))
print(f"\nZoom minimum near t ≈ {float(ts_zoom[min_idx]):.6f}, value ≈ {float(mags_zoom[min_idx]):.6e}")

# 9) Plot the zoomed region
plt.figure()
plt.plot(ts_zoom, mags_zoom)
plt.xlabel('t')
plt.ylabel(r'$|\Phi_{J_1}(0.5 + i\,t)|$')
plt.title('Zoom near first expected zero of  $\Phi_{J_1}$ ')
plt.grid(True)
plt.show()
```



Coarse minima (t , $|\Phi|$):

$t = 4.000000$, $|\Phi| = 1.769323e+00$
 $t = 14.800000$, $|\Phi| = 3.836504e-01$
 $t = 21.700000$, $|\Phi| = 4.351359e-01$
 $t = 25.700000$, $|\Phi| = 4.649122e-01$

Zoom minimum near $t \approx 14.300000$, value $\approx 7.752914e-01$

